SIMATS ENGINEERING

CO2

ASSESSMENT TOOL 3 – DRY RUN EVALUATION

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

NAME:THUVIKSHA V

REG NO:1924260035

COURSE NAME:ARTIFICIAL INTELLIGENCE

COURSE CODE:CSA1710

SUBMITTED TO:L.T.HERLIN

Annexure A – Dry Run Evaluation

Code of Conduct:

I, Thuviksha V (Register No. 1924260035), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:THUVIKSHA V

Annexure A – Dry Run Evaluation

1. Question 1 – Breadth-First Search (BFS)

Graph: A is connected to B and C; B is connected to D and E; C is connected to F and G. Visit children from left to right.

Iteration	Queue after adding children / before next visit	Visited Node
1	A	A
2	B, C	B
3	C, D, E	C
4	D, E, F, G	D
5	E, F, G	E
6	F, G	F
7	G	G

1. BFS traversal order: A → B → C → D → E → F → G.

2. Queue contents after each iteration:

- After visiting A: [B, C]
- After visiting B: [C, D, E]
- After visiting C: [D, E, F, G]
- After visiting D: [E, F, G]
- After visiting E: [F, G]
- After visiting F: [G]
- After visiting G: []

2. Question 2 – Depth-First Search (DFS)

Using left-to-right order, DFS goes as deep as possible before backtracking.

DFS traversal order: A → B → D → E → C → F → G.

Stack/visit trace: A, B, D; backtrack to B and visit E; backtrack to A and visit C, then F, then G.

3. Question 3 – 8-Puzzle using Breadth-First Search (BFS)

Initial state:

1 3 6
5 0 2
4 7 8

Goal state:

1 2 3
4 5 6
7 8 0

For the dry-run table, S0 is the initial state. The queue is shown using state labels to keep the table readable; the queue is FIFO. Neighbour moves are generated in the order Up, Down, Left, Right.

Iteration	Current State (0 is blank)	Queue Before Expansion	Queue After Expansion
1	S0: 1 3 6 / 5 0 2 / 4 7 8	[S0]	[S1, S2, S3, S4]
2	S1: 1 3 6 / 5 2 0 / 4 7 8	[S1, S2, S3, S4]	[S2, S3, S4, S5, S6]
3	S2: 1 3 0 / 5 2 6 / 4 7 8	[S2, S3, S4, S5, S6]	[S3, S4, S5, S6, S7, S8]
4	S3: 1 3 6 / 0 5 2 / 4 7 8	[S3, S4, S5, S6, S7, S8]	[S4, S5, S6, S7, S8, S9, S10]
5	S4: 1 3 6 / 5 7 2 / 4 0 8	[S4, S5, S6, S7, S8, S9, S10]	[S5, S6, S7, S8, S9, S10, S11, S12]

Answers to the questions

1. Node expanded in each of the first five iterations: S0, S1, S2, S3, and S4, respectively.

2. Using the stated Up, Down, Left, Right neighbour order and avoiding repeated states, 500 unique states are discovered by the time the goal is reached. The goal is dequeued as the 311th expanded state. Counts can vary if a different neighbour order or duplicate-handling convention is used.

3. One shortest solution path (moves of the blank tile): Right → Up → Left → Down → Left → Down → Right → Right.

State path:

Step	Puzzle State (rows separated by /)
S0 (Initial)	1 3 6 / 5 0 2 / 4 7 8
1	1 3 6 / 5 2 0 / 4 7 8
2	1 3 0 / 5 2 6 / 4 7 8
3	1 0 3 / 5 2 6 / 4 7 8
4	1 2 3 / 5 0 6 / 4 7 8
5	1 2 3 / 0 5 6 / 4 7 8
6	1 2 3 / 4 5 6 / 0 7 8
7	1 2 3 / 4 5 6 / 7 0 8
Goal	1 2 3 / 4 5 6 / 7 8 0

4. BFS finds the shortest solution in an unweighted 8-puzzle because it explores states level by level. Each move has equal cost, so the first time the goal is reached, it has the minimum number of moves.

Student Name: Thuviksha V Register No.: 1924260035